Data Mining & Business Intelligence (IT647)
Selected Topics In Data Science (IT688)

Information Technology Department
College Of Computer
Qassim University

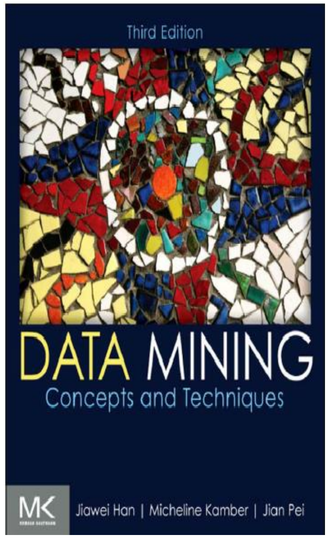
DR.MONA ALSALEEM

Data Mining & Business Intelligence Course_IT674

Data Mining in BI

— Chapter 4 —

Data Warehousing and Online Analytical Processing

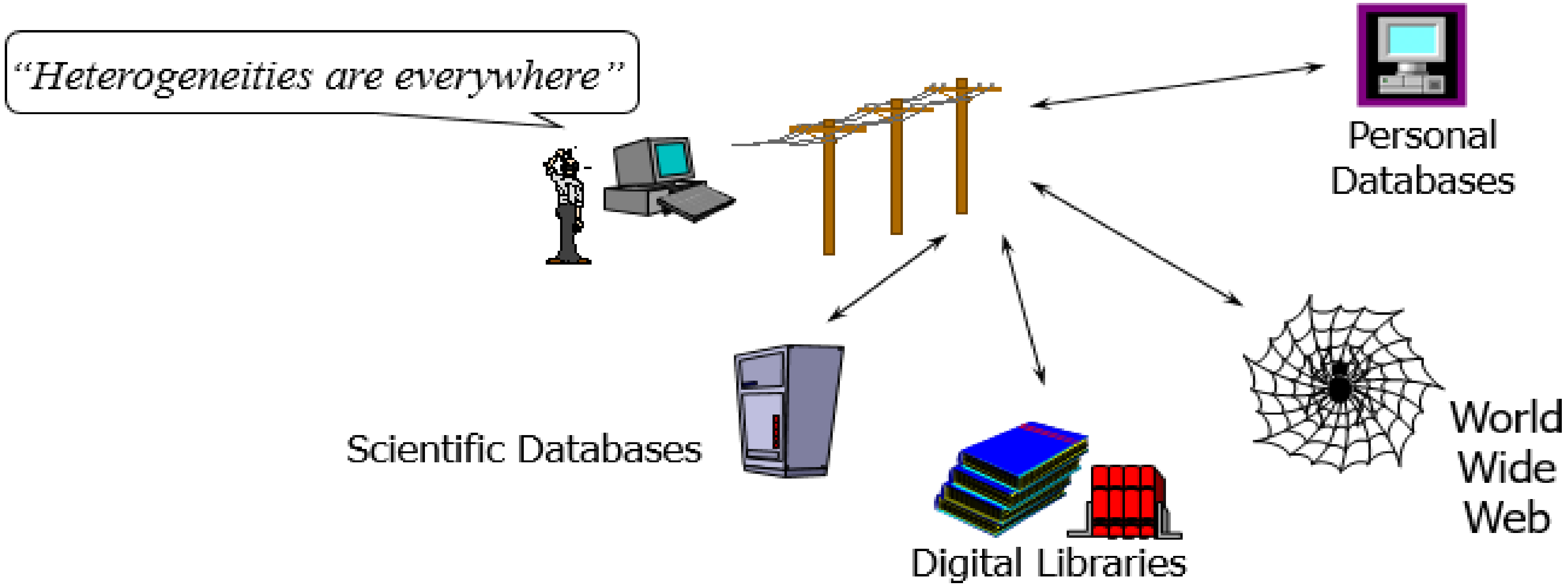


Text Book: “Data Mining: Concepts and Techniques”, Third Edition

Learning Objectives

- Why data warehouse
- What’s data warehouse
- What’s difference between OLAP and OLTP
- Data Warehouse Architecture
- Data Mart
- What’s multi-dimensional data model
- Summary

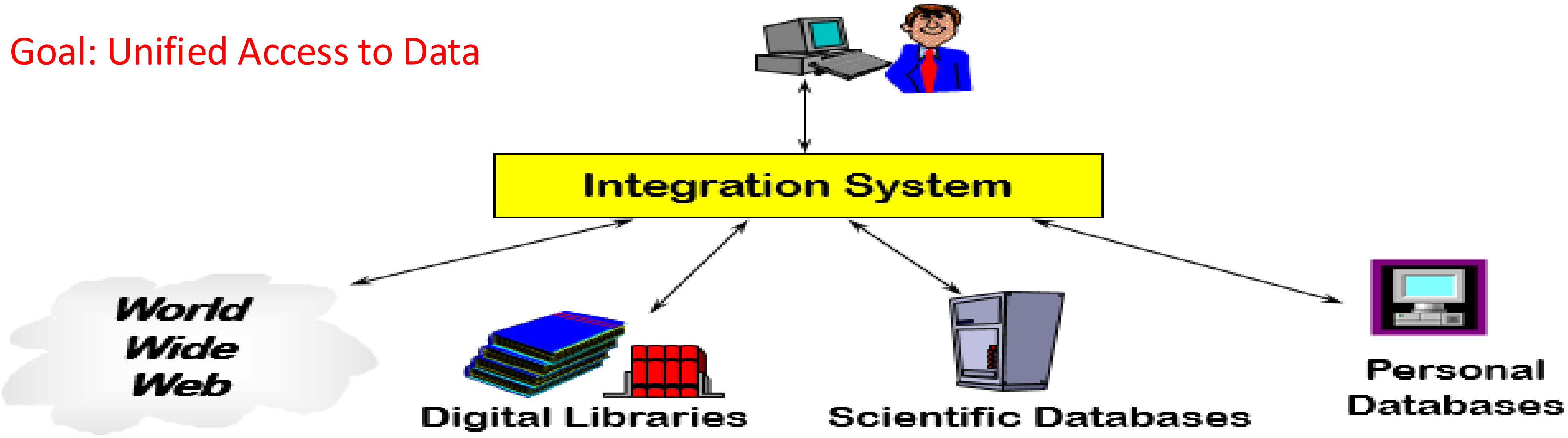
Introduction



- Different interfaces
- Different data representations
- Duplicate and inconsistent information

Introduction

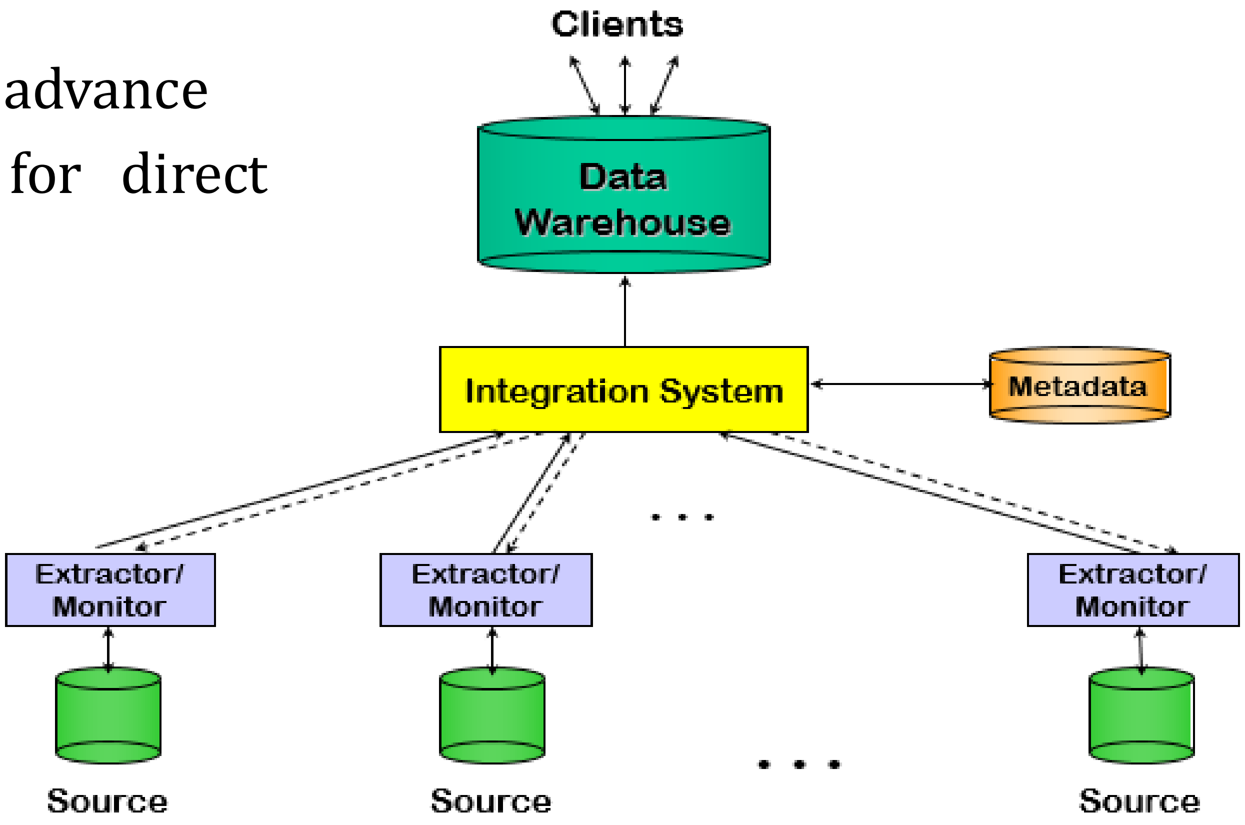
Goal: Unified Access to Data



- Collects and combines information
- Provides integrated view, uniform user interface
- Supports sharing

The Warehousing approach

- Information integrated in advance
- Stored in warehousing for direct querying and analysis



What is Data Warehouse

"A data warehouse is simply a single, complete, and consistent store of data obtained from a variety of sources and made available to end users in a way they can understand and use it in a business context."

Barry Devlin, *IBM Consultant*

A Data Warehouse is …

- Stored collection of diverse data
  - A solution to data integration problem
  - Single repository of information
- Subject-oriented
  - Organized by subject, not by application
  - Used for analysis, data mining, etc.

What is Data Warehouse

"A DW is a

- subject-oriented,
- integrated,
- time-variant,
- nonvolatile

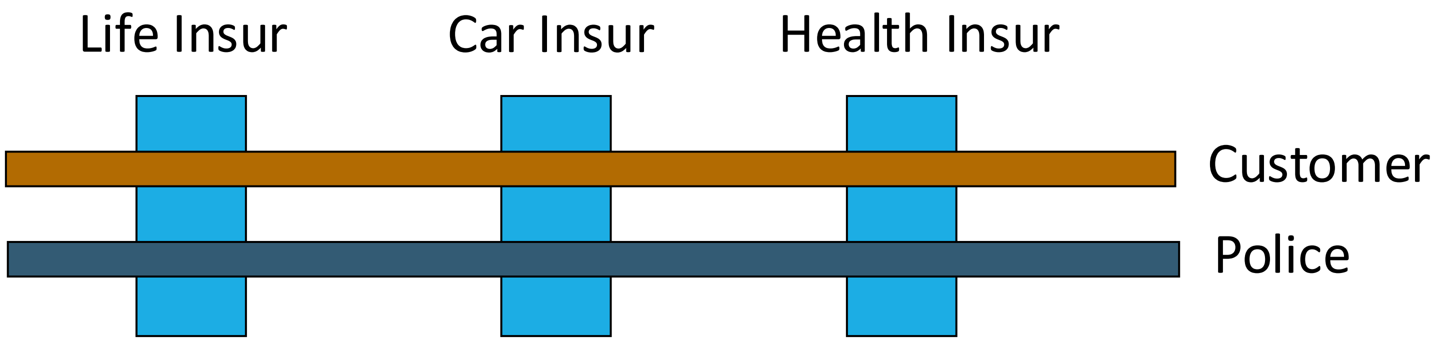
collection of data in support of management's decision making process."

William.H. Inmon, Building the Data Warehouse, 1996

Data Warehouse: Characteristics

Subject-oriented data:

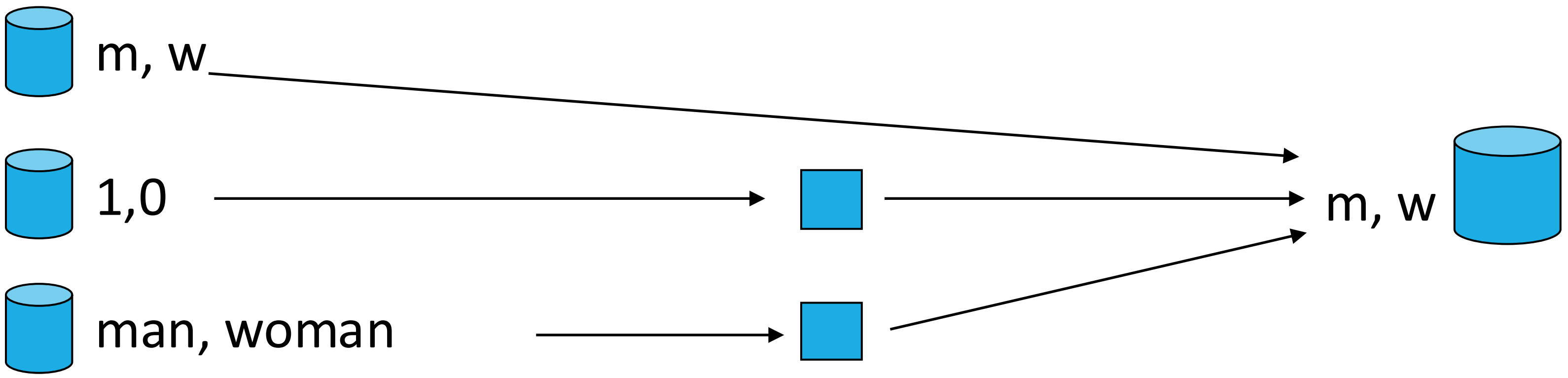
- A data warehouse is organized around major subjects such as customer, supplier, product, and sales.
- A data warehouse focuses on the modeling and analysis of data for decision makers.
- It provides a simple and concise view of particular subject issues by excluding data that are not useful in the decision support process.



Data Warehouse: Characteristics

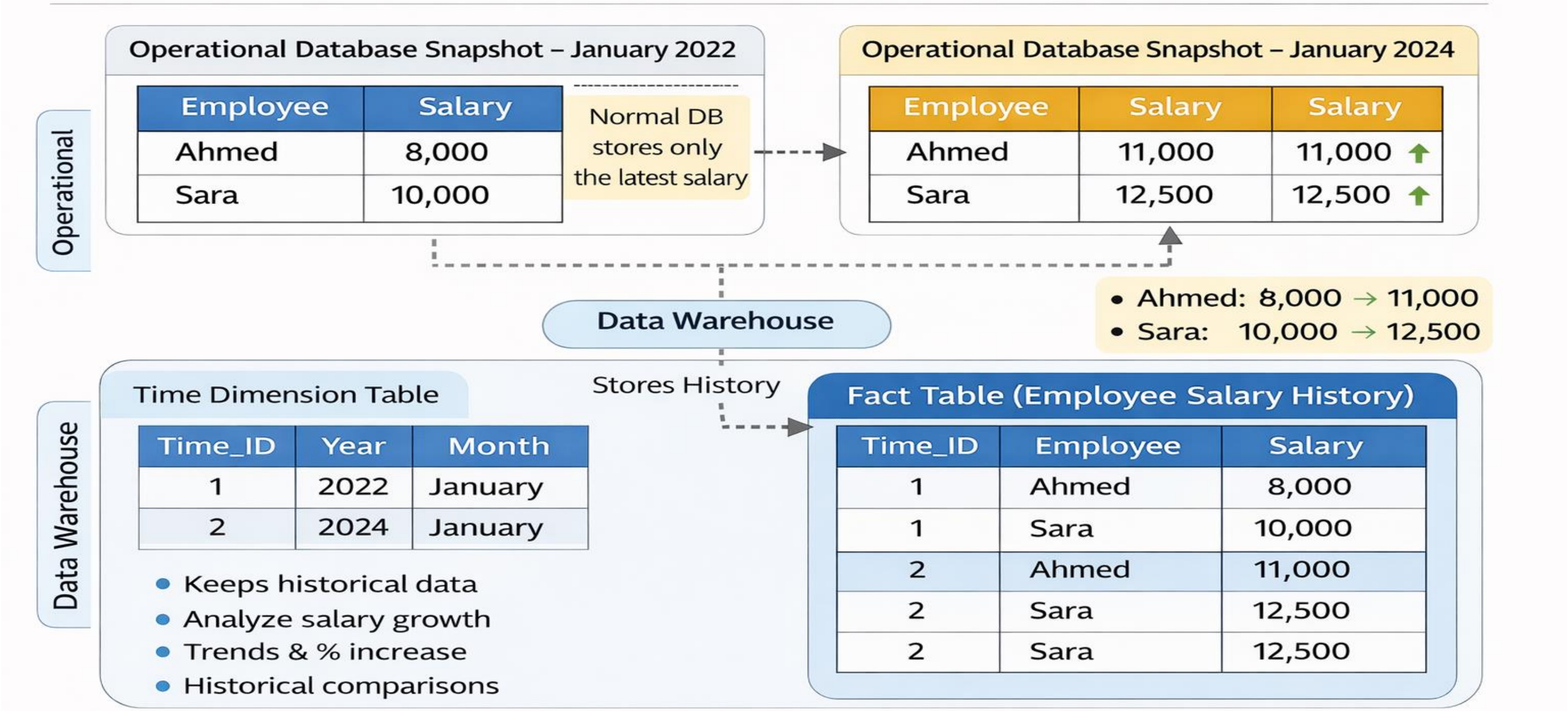
Integrated data:

- A data warehouse is usually constructed by integrating multiple heterogeneous sources, such as relational databases, flat files, and online transaction records.
- Data cleaning and data integration techniques are applied to ensure consistency in naming conventions, encoding structures, attribute measures.



Data Warehouse: Characteristics

- **Time-variant data:** Data are stored to provide information from an historic perspective (e.g., the past 5–10 years). Every key structure in the data warehouse contains, either implicitly or explicitly, a time element.

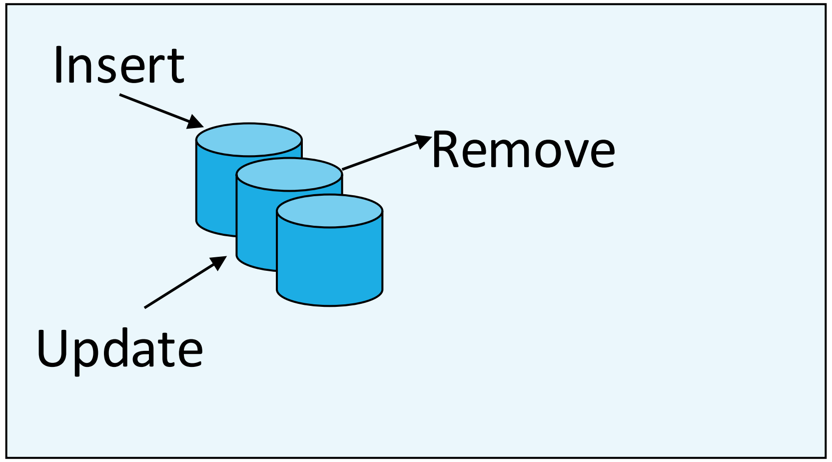


Data Warehouse: Characteristics

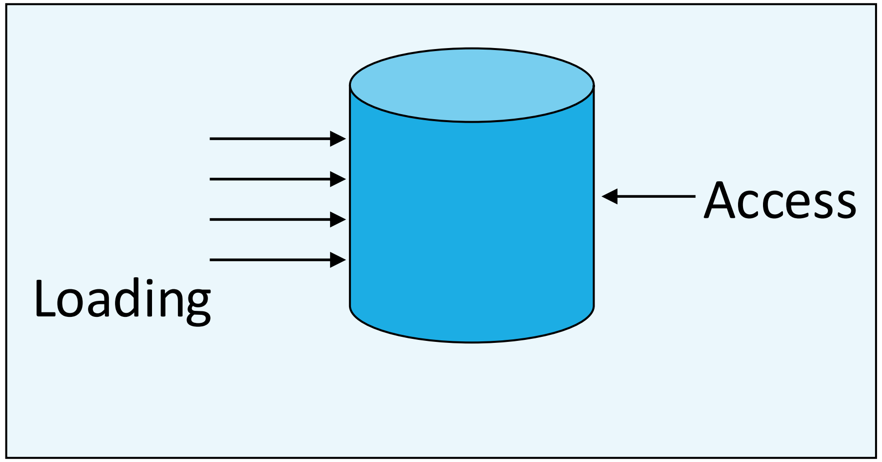
Nonvolatile data:

A data warehouse is always a physically separate store of data transformed from the application data found in the operational environment. Due to this separation, a data warehouse does not require transaction processing. It usually requires only two operations in data accessing: initial loading of data and access of data.

Database



Data Warehouse

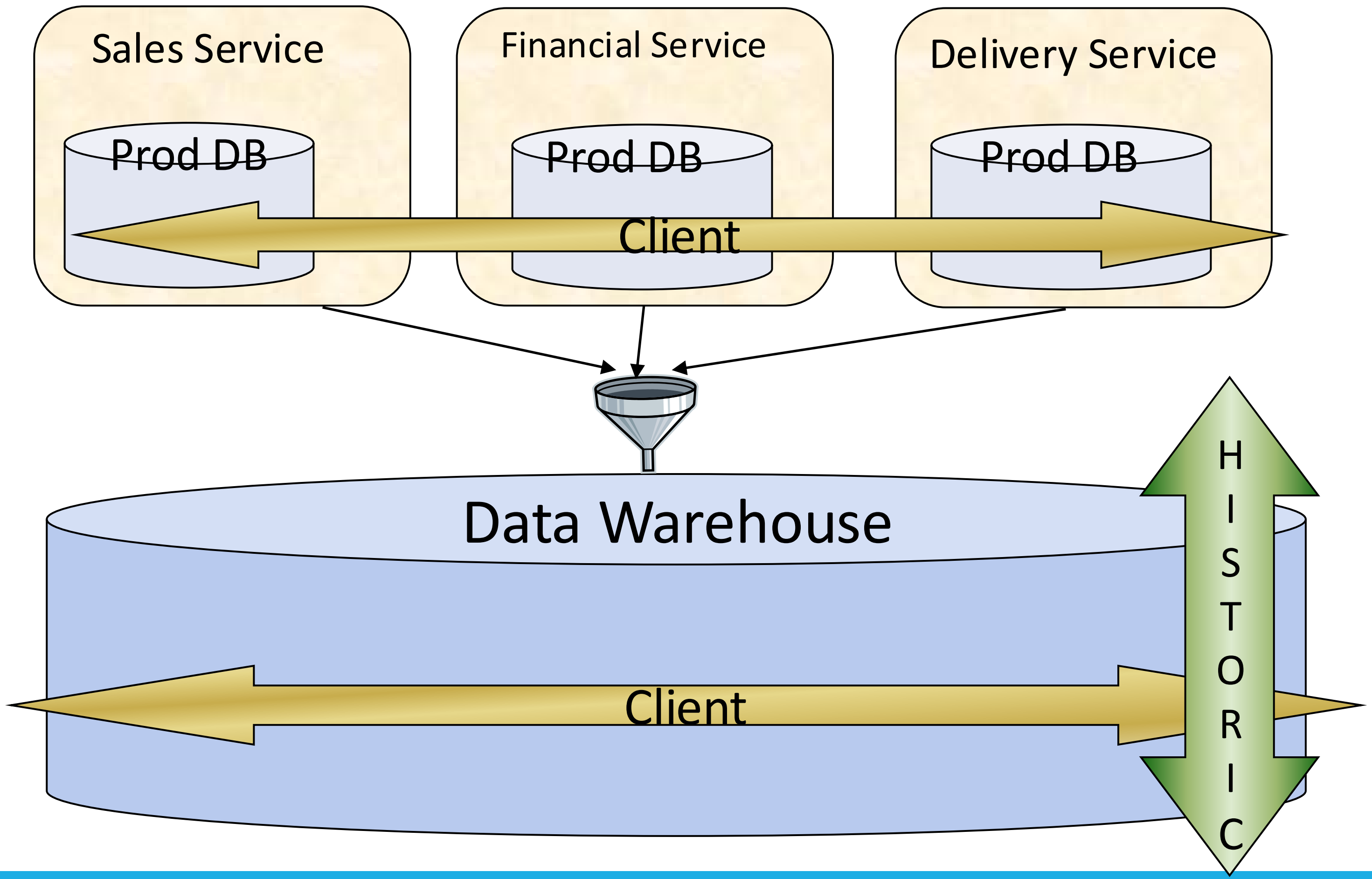


Data Warehouse: Applications

- Banking
- Health care
- Business,
- Logistics,
- Insurance,
- etc

DW vs DBMS

OLTP: On-Line Transactional Processing



OLAP: On-Line Analytical Processing

DW vs DBMS

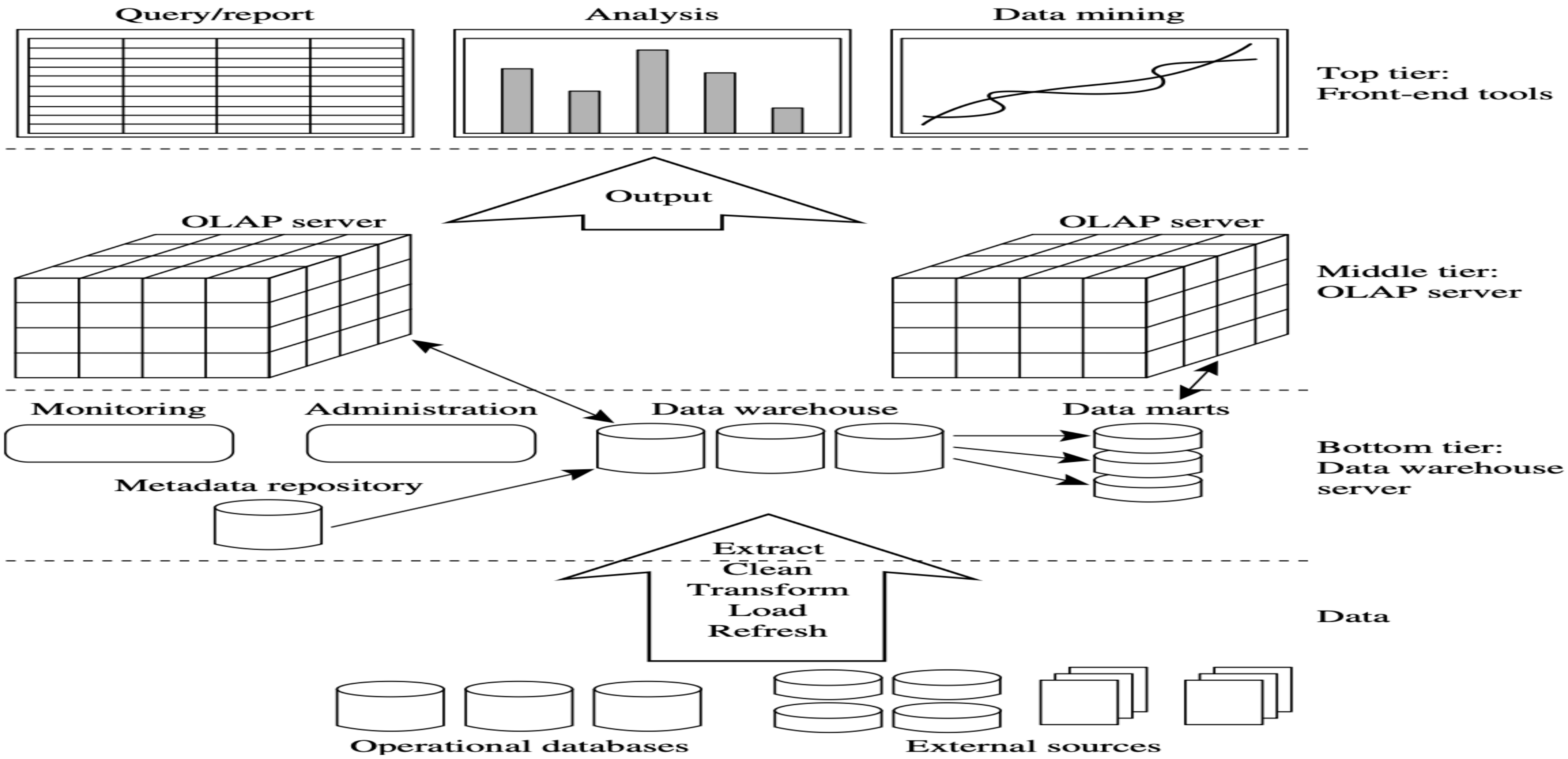
Characteristics	OLTP	OLAP
Use	DBMS	DW
Typical operation	Update	Analysis
Access type	Read/ Write	Read
Analysis level	Elementary	Global
Amount of information exchanged	Low	large
Orientation	line	multidimensional
Size	Low (GB)	Large (TB)
Seniority of data	current	Current/Historical

DW vs DBMS

- OLTP Systems are used to "run" a business

- The Data Warehouse helps to "optimize" the business

Data Warehouse Architecture



A three-tier data warehousing architecture

Data Warehouse Architecture

1. **The bottom tier** is a warehouse database server that is almost always a relational database system. Back-end tools and utilities are used to feed data into the bottom tier from operational databases or other external sources (e.g., customer profile information provided by external consultants).

2. **The middle tier** is an OLAP server that is typically implemented using either :

- A relational OLAP (ROLAP) model or
- A multi dimensional OLAP (MOLAP) model .

3. The top tier is a front-end client layer, which contains query and reporting tools, analysis tools, and/or data mining tools

Data Warehouse Architecture

ETL: definition

- Type of data integration that refers to the three steps (extract, transform, load)
- Used to blend data from multiple sources
- Discover, analyze and extract data from heterogeneous sources
- Allows cleaning and standardizing data
- Load data into a warehouse

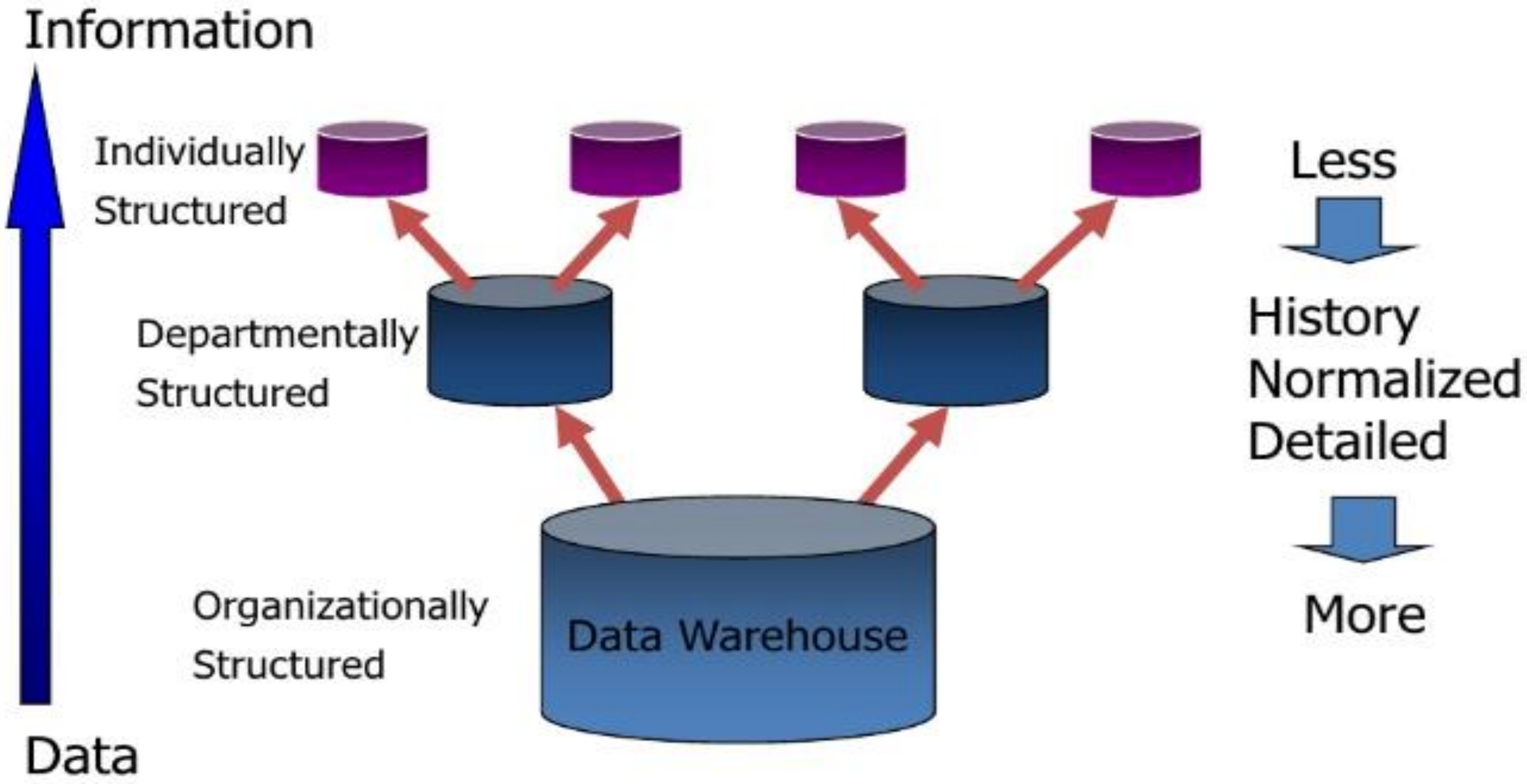
Data Marts

- A data mart is a scaled down version of a data warehouse that focuses on a particular subject area.
- A data mart is a subset of an organizational data store, usually oriented to a specific purpose or major data subject, that may be distributed to support business needs.

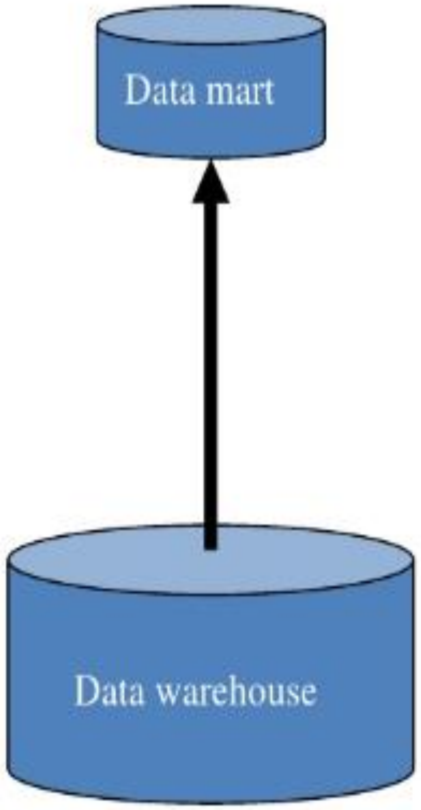
Reasons for creating a Data Mart

- Easy access to frequently needed data
- Creates collective view by a group of users
- Improves end-user response time
- Ease of creation in less time
- Lower cost than implementing a full Data Warehouse

From the Data Warehouse to Data Mart



Characteristics of the Departmental Data Mart



- Small
- Flexible
- Customized by Department
- OLAP
- Source in departmentally structured data warehouse

Data Warehouse: Dimensional Modeling

- **Dimensional modeling** (DM) names a set of techniques and concepts used in data warehouse design.

- Dimensional modeling is one of the methods of data modeling, that help us to store data in such way that it is relatively easy to retrieve data from the database.

Data Warehouse: Dimensional Modeling

- New types of table
  - Fact table
  - Dimension table
- New models
  - Star model
  - Snowflake model
  - Fact Constellations

Data Warehouse: Dimensional Modeling

- There are two main types of objects in a dimensional model
  - **Facts** are quantitative measures that we wish to analyze and report on
  - **Dimensions** contain textual descriptors of the business. They provide context for the facts

Data Warehouse: Dimensional Modeling

Fact Tables

- A fact table stores quantitative information for analysis
- Contains two or more foreign keys
- Tend to have huge numbers of records
- Useful facts tend to be numeric and additive

Data Warehouse: Dimensional Modeling

Example: Fact Table

Suppose that a company sells products to customers. Every sale is a fact that happens, and the fact table is used to record these facts. For example:

Time ID	Product ID	Customer ID	Unit Sold
4	17	2	1
8	21	3	2
8	4	1	1
5	20	2	5
3	4	4	7

Data Warehouse: Dimensional Modeling

Dimension Table

- A dimension table stores attributes, or dimensions, that describe the objects in a fact table.
- A data warehouse organizes descriptive attributes as columns in dimension tables.
- For example: a customer dimension’s attributes could include first and last name, birth date, gender, etc., or a website dimension would include site name and URL attributes.
- A dimension table has a primary key column that uniquely identifies each dimension record (row).

Data Warehouse: Dimensional Modeling

Example: Dimension Table

Customer ID	Name	Gender	Income	Education	Region
1	Brian Edge	M	2	3	4
2	Fred Smith	M	3	5	1
3	Sally Jones	F	1	7	3

Data Warehouse: Dimensional Modeling

Dimension Table Granularity

- A dimension contains hierarchically organized members
  - Each member belongs at a particular hierarchical level (or granularity level)
  - Dimension granularity: number of hierarchical levels
- Example: Time dimension
  - year - semester - trimester - month

Data Warehouse: Dimensional Modeling

Evolution of Dimensions

- **Slowly changing dimensions**
  - A client can get married, have children, etc.
  - A product may change names
  - Situation management, 3 solutions:
    - Overwrite the existing data value
    - Adding a new record
    - Adding a new attribute

Data Warehouse: Dimensional Modeling

Slowly changing dimensions: Overwrite the existing data value

- Correction of erroneous information
- Advantage : Easy to implement
- Disadvantages: Loss of trace of the previous attribute values

Prod key	Prod Description	Prod Group
12345	Intelli-Kids	Soft

Educational games

Data Warehouse: Dimensional Modeling

Slowly changing dimensions: Adding a new record

- Using a substitution key
- Advantage : Track the evolution of attributes
- Disadvantages: Increases the volume of the table

Prod key	Prod Description	Prod Group
12345	Intelli-Kids	Soft
25963	Intelli-Kids	Educational games

Data Warehouse: Dimensional Modeling

Slowly changing dimensions: Adding a new attribute

- Origin value / current value
- Advantage : Having two simultaneous visions of the data
- Disadvantages: Unsuitable for follow several intermediate attribute values

Prod key	Prod Description	Prod Group	New group
12345	Intelli-Kids	Soft	Educational games

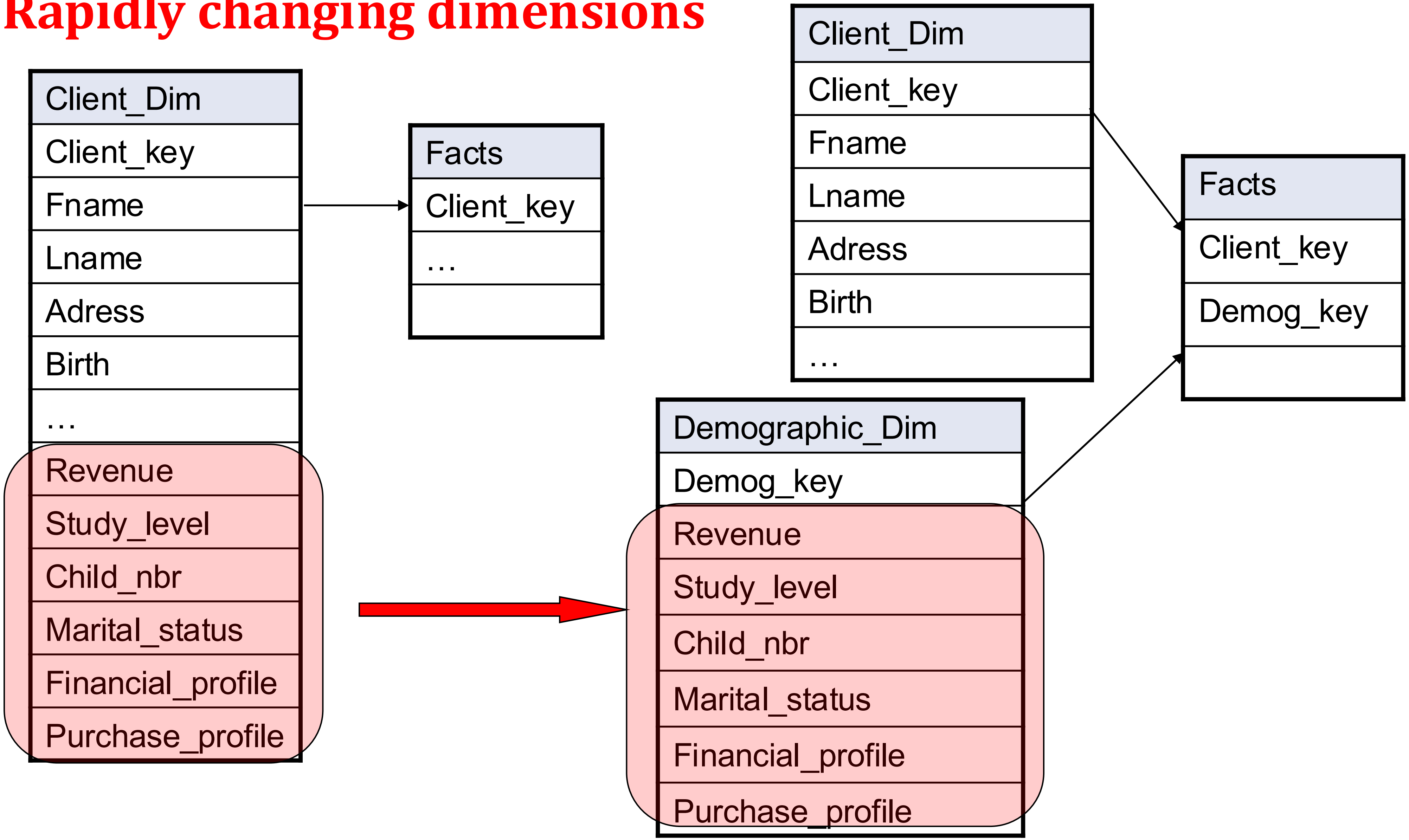
Data Warehouse: Dimensional Modeling

Evolution of Dimensions

- **Rapidly changing dimensions**
  - Very frequent changes (every month) which we want to preserve the history.
  - Solution: isolate the attributes rapidly changing

Data Warehouse: Dimensional Modeling

Rapidly changing dimensions



Data Warehouse: Dimensional Modeling

Facts and Dimensions

Criteria	Fact Attributes	Dimension Attributes
Purpose	Measurements for reporting or analysis	Constraints or qualifiers for the measurements
Data type	Additive or semi-additive quantitative data	Textual, descriptive
size	Large number of records	Smaller number of records

Data Warehouse: Dimensional Modeling

Conceptual Modeling of Data Warehouse

- Star Model
- Snowflake Model
- Fact Constellations

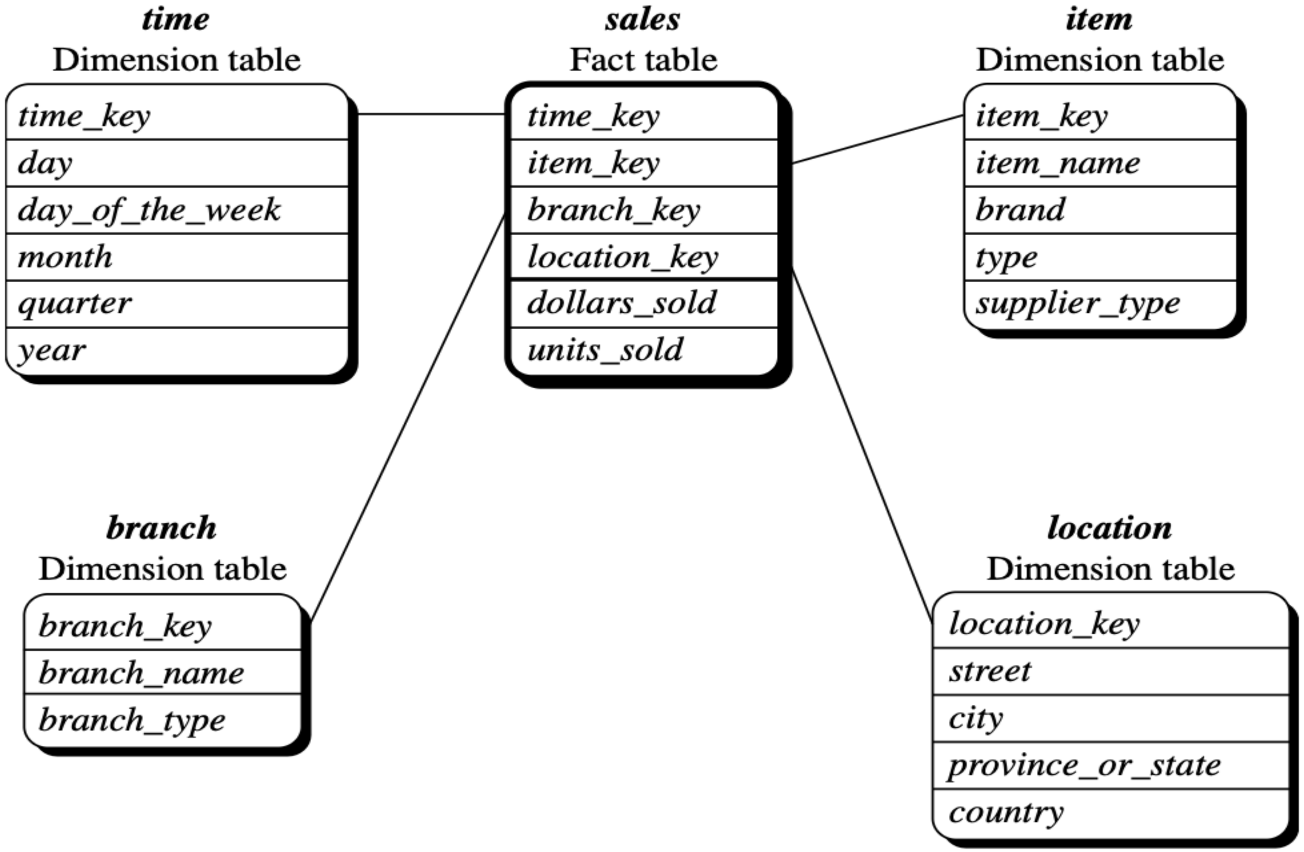
Data Warehouse: Dimensional Modeling

Star Model

- The star schema architecture is the simplest data warehouse schema.
- It is called a star because the diagram resembles a star, with points radiating from a center.
- The center of the star consists of fact table and the points of the star are the dimension tables

Data Warehouse: Dimensional Modeling

Star Model



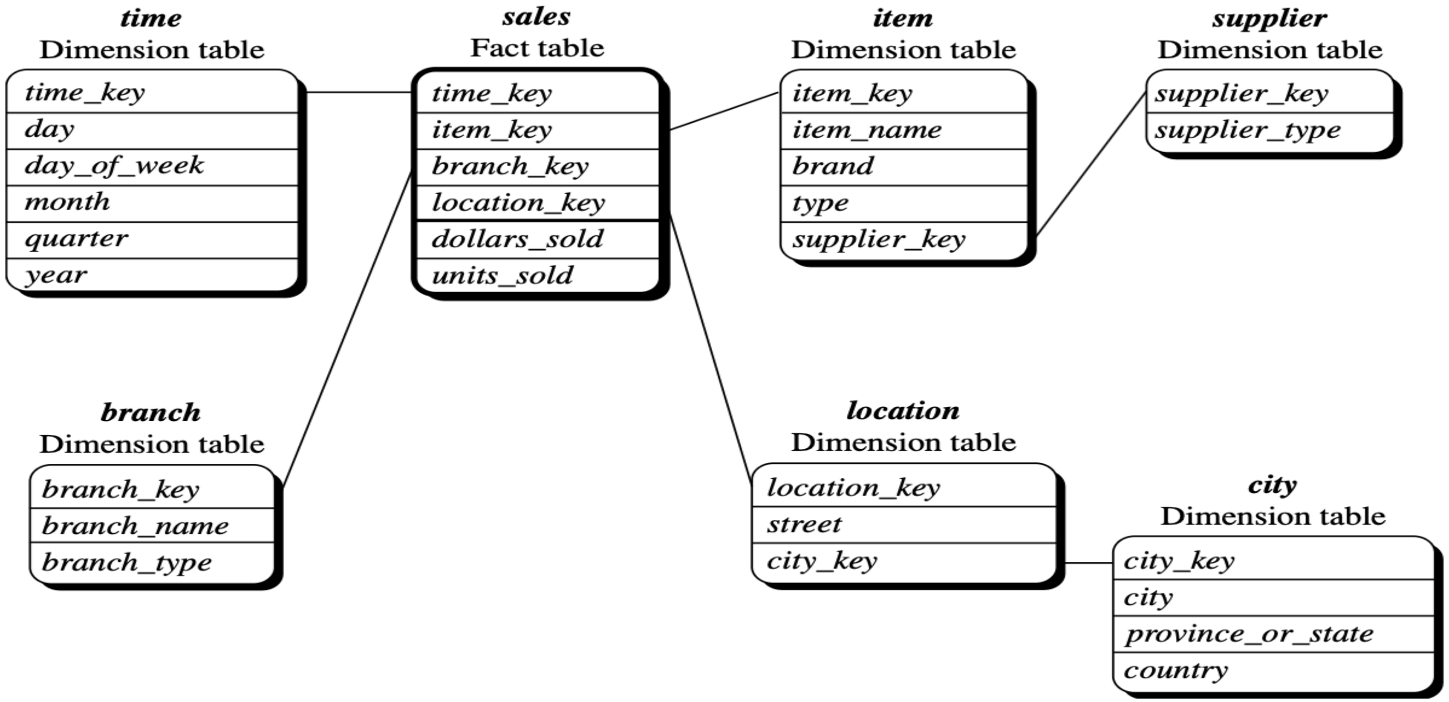
Data Warehouse: Dimensional Modeling

Snowflake Model

- Some dimension tables in the snowflake schema are normalized
- Unlike star schema, the dimensions table in a snowflake schema are normalized.
- For example, the item dimension table in star schema is normalized and split into two dimension tables, namely item and supplier table.

Data Warehouse: Dimensional Modeling

Snowflake Model



Data Warehouse: Dimensional Modeling

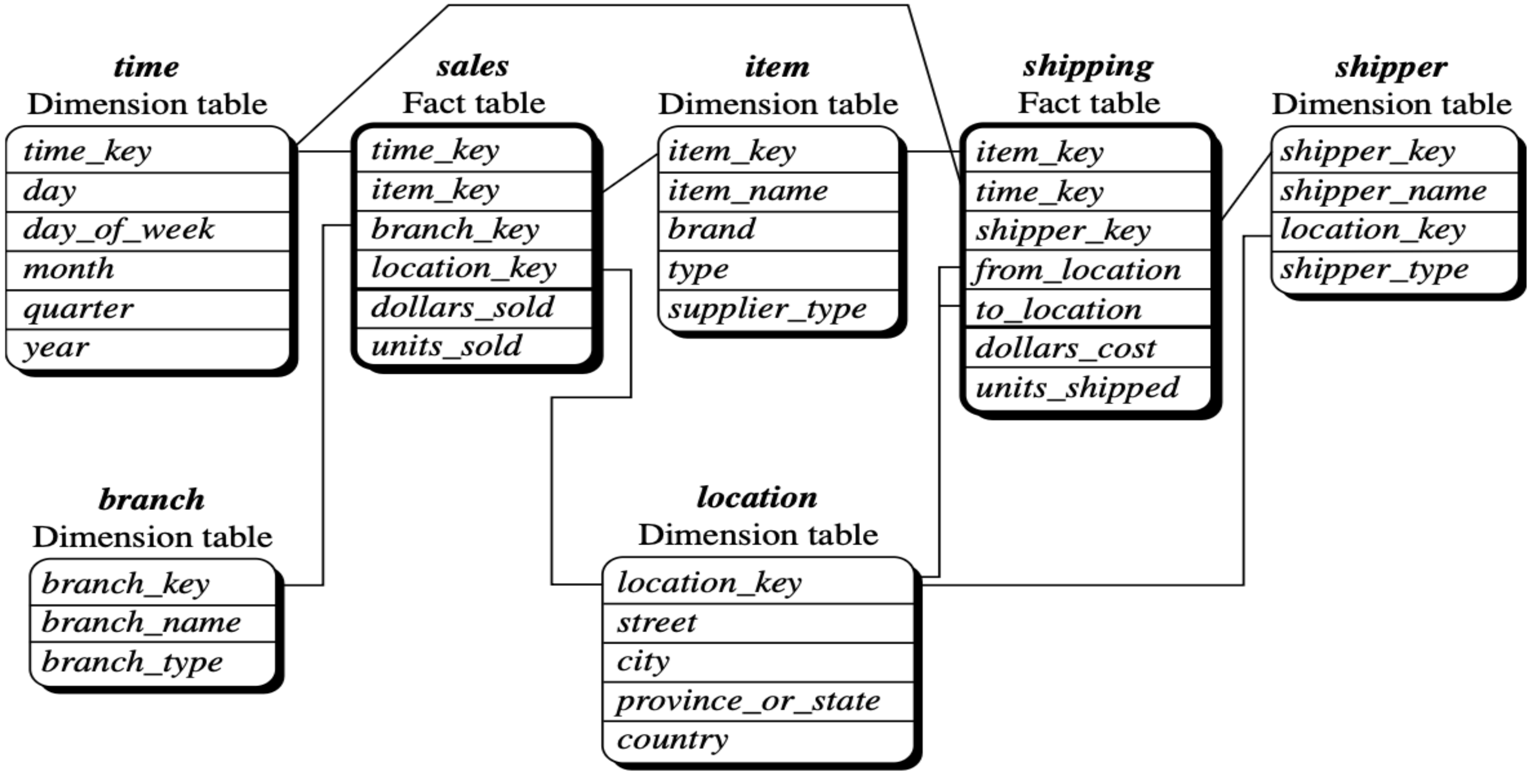
Fact Constellation Model

- A fact constellation has multiple fact tables. It is also known as galaxy schema

- It is possible to share dimension tables between fact tables. For example, time, item and location dimension tables are shared between the sales and shipping fact table

Data Warehouse: Dimensional Modeling

Fact Constellation Model



Multi-Dimensional Databases

A multi**dimensional** database (MDDB) is a computer software system designed to allow for the efficient and convenient storage and retrieval of large volumes of data that are

(1) intimately related and

(2) stored, viewed and analyzed from different **perspectives**. These perspectives are called **dimensions**.

Multi-Dimensional Databases

Cube

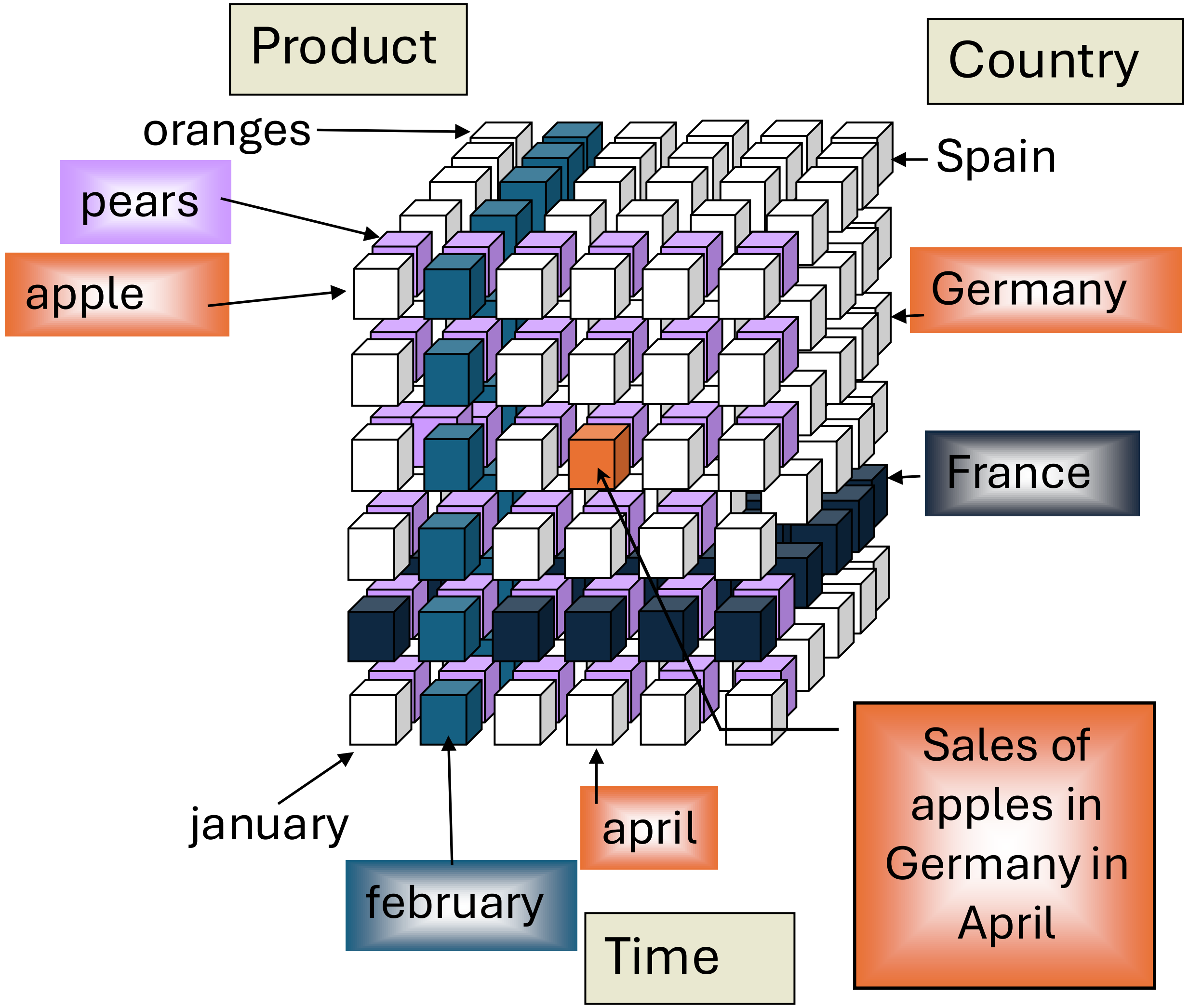
- Multidimensional data modeling that facilitates the analysis of a quantity in different dimensions
  - Time
  - Geographical location
  - Etc.

Example

Product	
PK	id_prod
	name
	group

Purchase	
PK	id_purch
FK	id_cust
	id_prod
	Quantity

Customer	
PK	id_cust
	name
	adress



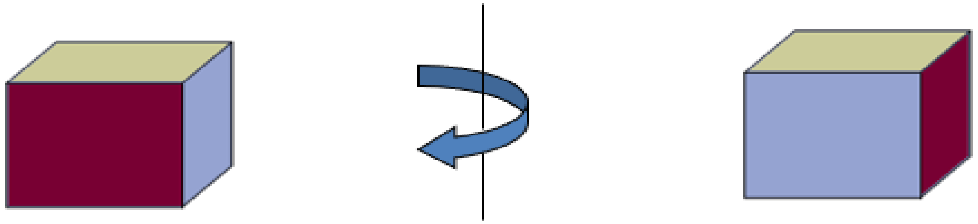
What Can We Do With a Cube? Typical OLAP Operations

Pivot (Rotate)

- is a visualization operation that rotates the data axes in view to provide an alternative data presentation
- present another face of the cube

	05	06	07
Egg	445	510	524
meat	350	326	329

	05	06	07
Idf	383	417	429
Ain	412	419	424



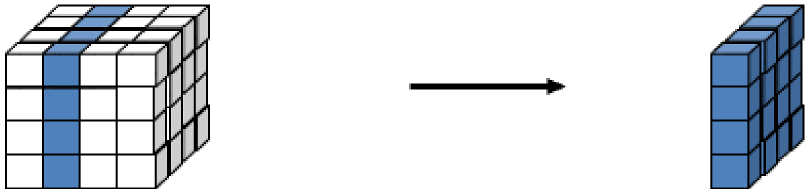
Typical OLAP Operations

Slicing

- **The slice operation performs a selection on one dimension of the given cube, resulting in a subcube.**
- **Work on a slice of the cube**

		05	06	07
egg	Idf	220	265	284
	Ain	225	245	240
meat	Idf	163	152	145
	Ain	187	174	184

		06
egg	Idf	265
	Ain	245
meat	Idf	152
	Ain	174



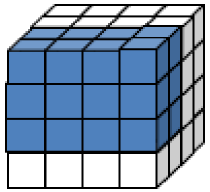
Typical OLAP Operations

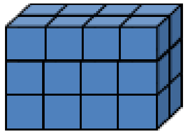
Dicing

- The dice operation defines a subcube by performing a selection on two or more dimensions.
- Work on a sub-cube

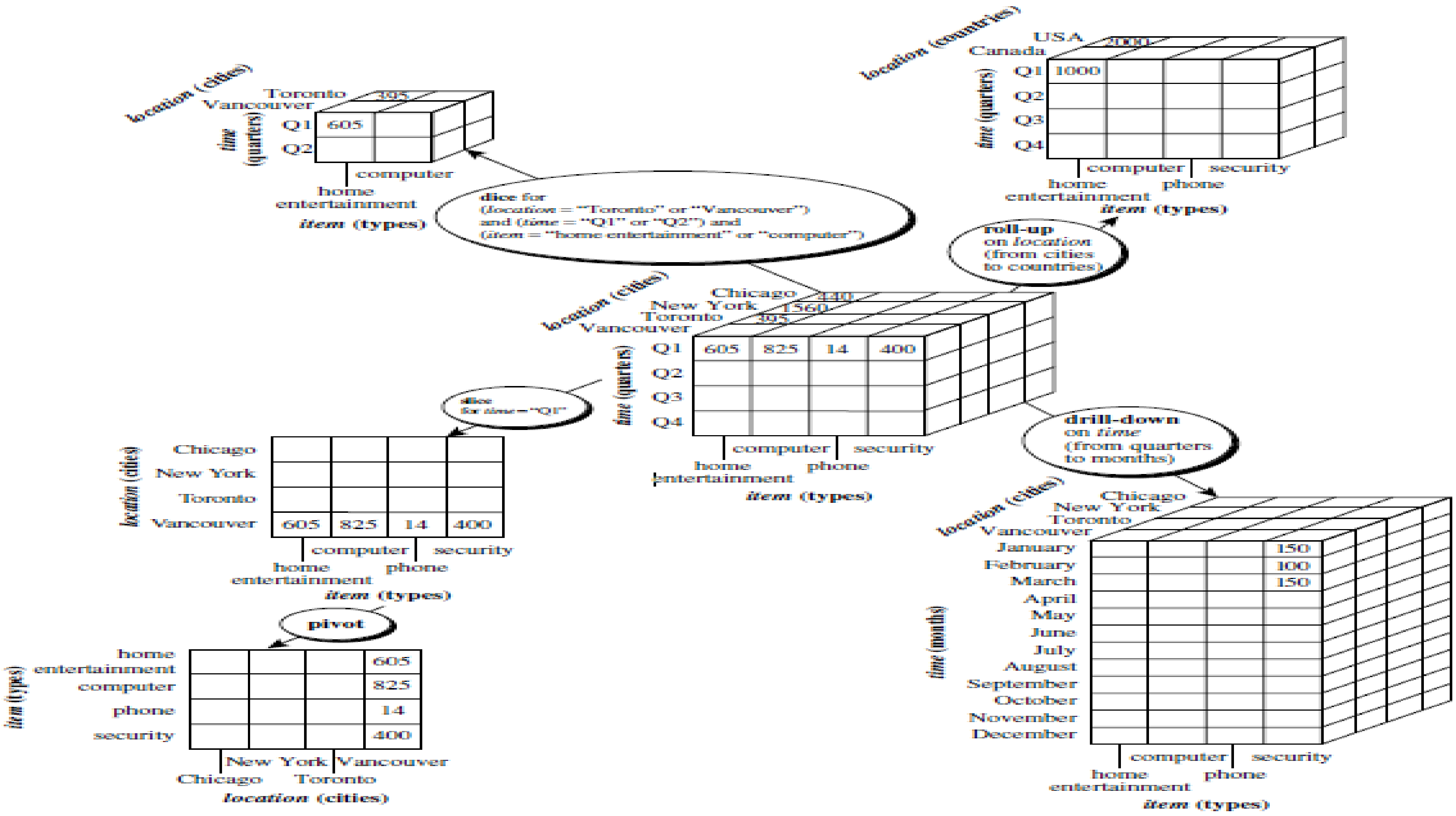
		05	06	07
egg	Idf	220	265	284
	Ain	225	245	240
meat	Idf	163	152	145
	Ain	187	174	184

		05	06	07
egg	Idf	220	265	284
	Ain	225	245	240





Typical OLAP Operations



Summary

- Data warehouse is a subject-oriented, integrated, time-varying, nonvolatile collection of data in support of management's decision making process.
- Difference between OLTP (On-Line Transactional Processing) and OLAP (On-Line Analytical Processing)
- Data mart is a scaled down version of a data warehouse that focuses on a particular subject area
- Dimensional modeling (DM) names a set of techniques and concepts used in data warehouse design.

Thank you